

# THOMAS TSANGOU

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## SUMMARY

Detail-oriented Computer Scientist and Data Engineer with hands-on experience in predictive modeling, automation, and infrastructure. Proven ability to take projects from conception to completion, including architecting on-premise cloud systems and developing machine learning models. Eager to apply skills in Python, Linux, and MLOps to build robust data solutions.

## EDUCATION

**Hunter College/CUNY** | New York, NY | Aug 2020-Dec 2025 *Bachelor of Arts, Computer Science, Minor in Mathematics*

**NYC College of Technology/CUNY** | Brooklyn, NY | Aug 2017-Aug 2020 *Associate of Applied Science, Computer Systems Technology*

## SKILLS

- **Data Science & Machine Learning:** Predictive Modeling, ETL, Data Pipelines, Feature Engineering, Exploratory Data Analysis (EDA), Data Quality Validation (GX Core)
- **Infrastructure & Networking:** Linux System Administration (Ubuntu), Reverse Proxy, DNS Management, VPN Configuration, SSH Hardening, Docker Orchestration, Apache Airflow (DAGs)
- **Programming & Databases:** Python (Pandas, Scikit-learn, NumPy), C++, SQL, Node.js, Next.js, Git version control, TypeScript, Bash, PostgreSQL
- **Tools:** Power BI, IBM Cloud, IBM Guardium, Streamlit, Microsoft Excel, IBM WatsonX

## EXPERIENCE

**IBM Corporation** | Data Engineering Intern | Security | Feb 2023-Jan 2024 - Automated onboarding of 10,000 databases into IBM Guardium using Python scripting and REST APIs, streamlining repetitive tasks - Developed a Power BI project to integrate Python Scripts for team-wide deployment, improving data processing and reporting efficiency - Enhanced data analysis capabilities by generating detailed reports with pandas, and automating Excel interactions using pywin32

**IBM Corporation** | Talent Acquisition Intern | Jun 2022-Dec 2022 - Collaborated with key stakeholders and a team of software developers to design, implement, and maintain new features for the Select for IBM (SFI) software. - Managed and optimized a Cloudant (NoSQL) Database using TypeScript, ensuring efficient data handling and robust error management

**Work Ed** | Program Lead | Jan 2024-Present - Developing and delivering lessons on AI concepts and chatbot development to over 100 students. - Translated complex technical requirements into engaging learning modules, coaching students on creating AI-powered chatbots with IBM WatsonX Assistant.

**Morrison Mentors** | Lead Instructor | Nov 2024-Present - Led STEAM education programs across four school locations, delivering engaging, hands-on classes to over 100 students in robotics, programming, and web design. - Guided students and Junior instructors in applying the scientific method to structure, visualize, and interpret data sets for hypothesis testing. - Acted as the primary point of contact for workshop logistics, managing equipment, materials, and learning environments for the team.

## LEADERSHIP & PROJECTS

**Headstarter AI** | Software Engineering Fellow | Jul 2024 - Present - Participating in a 7-week remote fellowship focused on building 5 AI projects and a capstone project

**Football Analytics Engine (FAE)** | Lead Developer & Architect | July 2024-Present - Architected a modular analytics platform using Docker to orchestrate PostgreSQL, Streamlit, and ETL workers, ensuring a scalable microservices environment for sports data processing. - Engineered automated ETL pipelines via Apache Airflow (DAGs), utilizing Selenium and Headless Chromium to extract multi-season player and squad metrics from FBRef while bypassing anti-bot measures. - Integrated MLOps best practices by enforcing data quality through GX Core (Great Expectations) and deploying predictive models to identify top-performing players and squads.

**On-Premise Private Cloud Infrastructure** | System Architect | Jun 2025-Present - Provisioned and hardened a bare-metal Ubuntu Server on workstation-class hardware (i7 vPro, 32GB RAM), implementing SSH key-based authentication to ensure strict access control. - Architected a secure remote access layer using Cloudflare Tunnels and Zero Trust Access policies to enforce identity-based authentication (OTP) for internal administrative tools. - Orchestrated containerized services via Docker and Portainer, specifically configuring Cockpit with custom WebSocket and CORS policies for secure remote server management. - Engineered a local AI development environment by bridging Ollama-hosted models (Qwen, Llama) to agentic IDEs like Cursor and Windsurf using custom OpenAI-compatible API endpoints.